

Meeting - Dobril Radoslavov

(Ministry of Defense)

Questions:

- 1) Може ли да ни даде контакти (основно в Украйна) да ни кажат от първо лице какви са им нуждите и следователно да разберем как да разработим дронове максимално в съответствие с техните нужди
- 2) Той има ли идея дали има реална нужда от такъв продукт?
- 3) Използват ли се логистични дронове в момента на бойното поле?
- 4) Crisis response → при пожари, земетресения, наводнения...
- ↳ Гражданска защита е към МВР, но въпреки всичко той има ли някаква информация?
- 5) С какви дронове разполагат в Украйна? Всичко ли е насочено към камикадзе и surveillance?
- 6) Какви дронове се разработват в момента в България?

Open questions:

- 1) What type of propellers are we going to use? Any CAD Files for them?
 - 2) What type of motors are we going to use?
 - 3) Currently, we are drawing the Full scale drone. Should we switch to the prototype already?
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- 1) Go through Abi's systems engineering and give feedback (html File in Discord).
- 2) $280 \times 240 \times 300$
depth $\times$ width $\times$ height } Martin's 3D printer
- 3) Start designing the prototype A/C

ORGO Meeting 15.02.2026

1)

1) Speed is everything nowadays! Look at companies in Asia. They are ahead of us because they are faster.

ORGOItly enables extremely fast deliveries for different companies!

2x 24V

↳ Battery clear

↳ Motor clear

~10 000 mAh

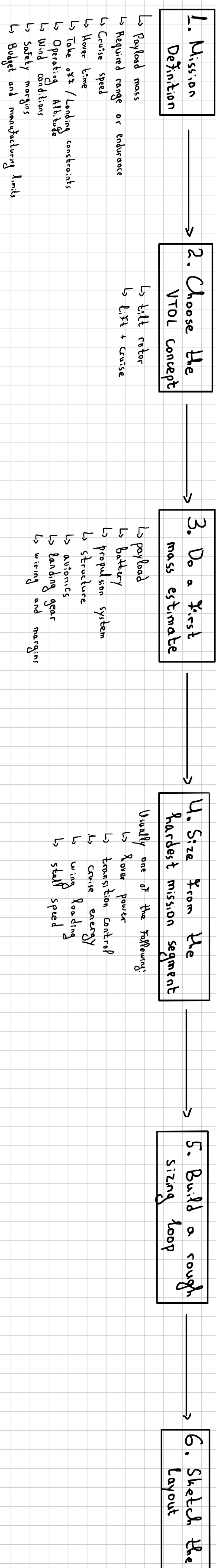
$\left\{ \begin{array}{l} 2_{prop} \rightarrow 18 - 22 \text{ inch} \\ 3_{prop} \rightarrow 10 - 13 \text{ inch} \end{array} \right.$

→ точен мотор, колко време издържа

↳ What props exactly?

↳ Control system?

- ↳ Define V-tail angle and airfoil type
- ↳ Engine pylons?



22.03.2026

→ kuno

→ V-tail

- Wingcopter (datasheet)

- Joby

- Quantum

- Skyways

- Zipline (V-tail)

$V_{cruise} \rightarrow 24 \text{ m/s}$

$V_{max} \rightarrow 42 \text{ m/s}$

$V_{stall} \rightarrow 12 \text{ m/s}$

$$L = \frac{1}{2} \rho \cdot V^2 \cdot S \cdot C_L = W$$

$$\Rightarrow S = \frac{2W}{\rho \cdot V_{stall}^2 \cdot C_{Lmax}}$$

$$W = 15 \cdot 9,80665 = 147,0998 \text{ N}$$

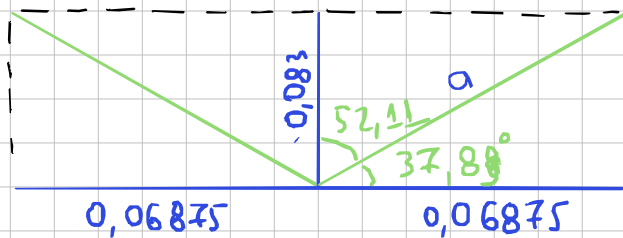
$$V_{stall} = 12 \text{ m/s}$$

$$C_{Lmax} \rightarrow \text{NACA 2412} \approx 1,4 \text{ as first approx.}$$

$Re \approx 500\,000$ at cruise speed and chord $0,3 \text{ m}$

$$\rho_{\infty} \approx 1,17$$

$$\Rightarrow S = \frac{2 \cdot 15 \cdot 9,80665}{1,17 \cdot 12^2 \cdot 1,4} = 1,247 \text{ m}^2$$

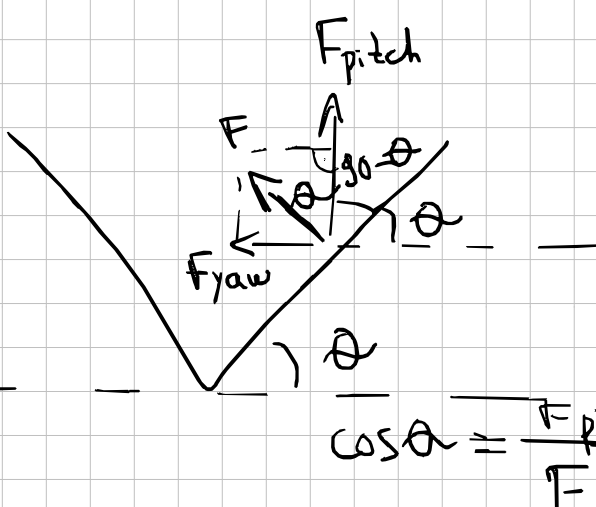


$$\cos 52,11 = \frac{0,083/2}{a}$$

$$a = \frac{0,0415}{\cos 52,11} = 0,0675$$

$$\cos 37,88 = \frac{0,06875}{a}$$

$$a = \frac{0,06875}{\cos 37,88} = 0,087$$



$$F_{pitch} = \cos \alpha \cdot F$$

$$F_{yaw} = \sin \alpha \cdot F$$

Workflow:

Wingspan bw
wing area S_w

⊕

choose: C_{Hr} / C_{Vr}
 L_{Hr} / L_{Vr}

calculate
 S_{Vr} / S_{Hr}

calculate
V-tail
angle
 α

You are
controlling
only this

calculate
V-tail
surface area
 S_{V-tail}

To Do:

CW 13

- 1) Add the workflow to the wing and V-tail design worksheet.
- 2) Find proper components and place them to evaluate the fuselage dimensions
- 3)

02.04.2026

$$\bar{X}_{CG} - \bar{X}_{AC} [\%] \rightarrow 5\% \dots 10\%$$

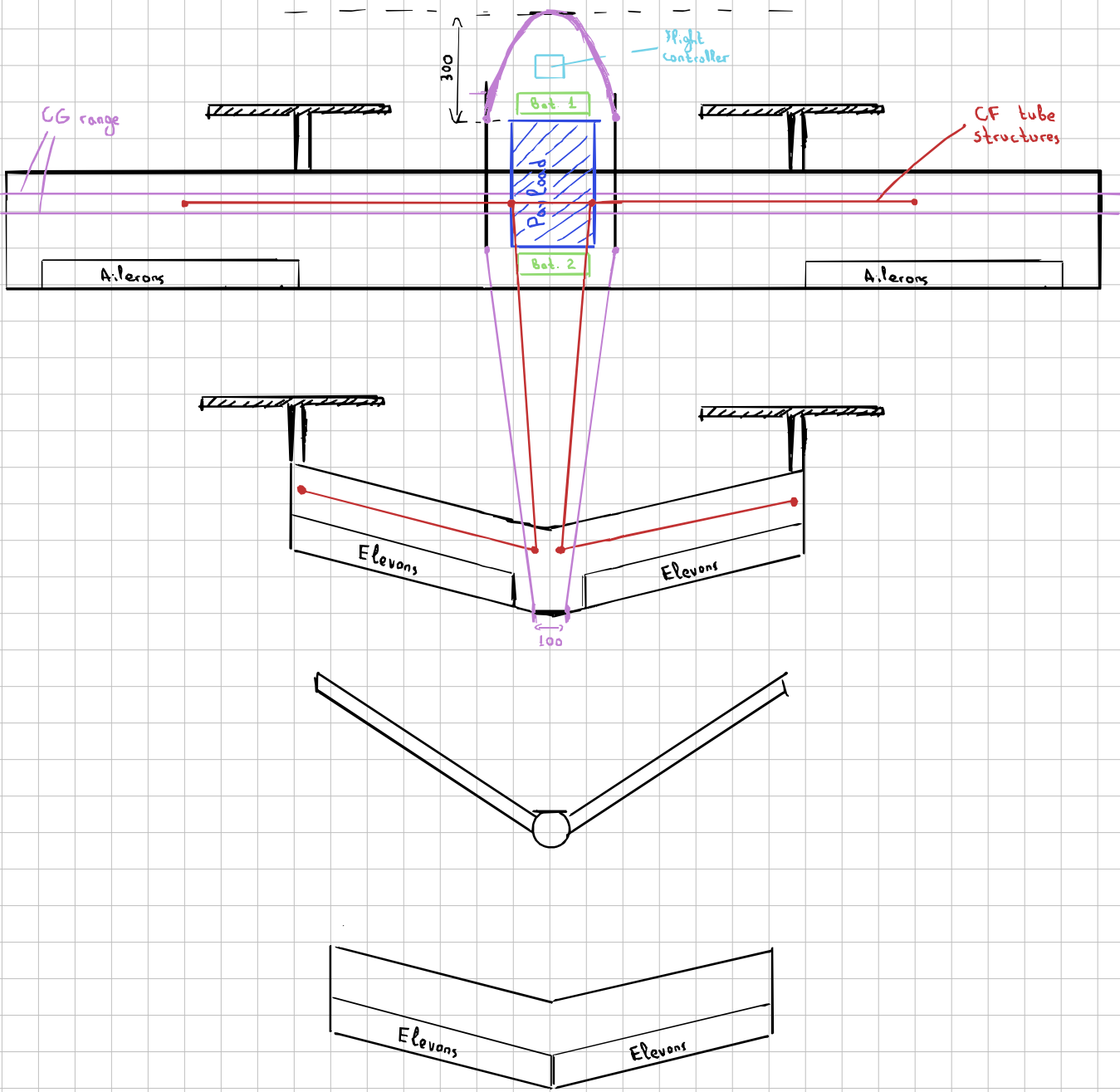
↳ For stable aircraft, the stability measure is negative

$$\bar{X}_{AC} = \frac{C_{Ma,w} - C_{Ma,fus} + C_{Ma,hr}}{C_{Ld,w} + C_{Ld,hr}}$$

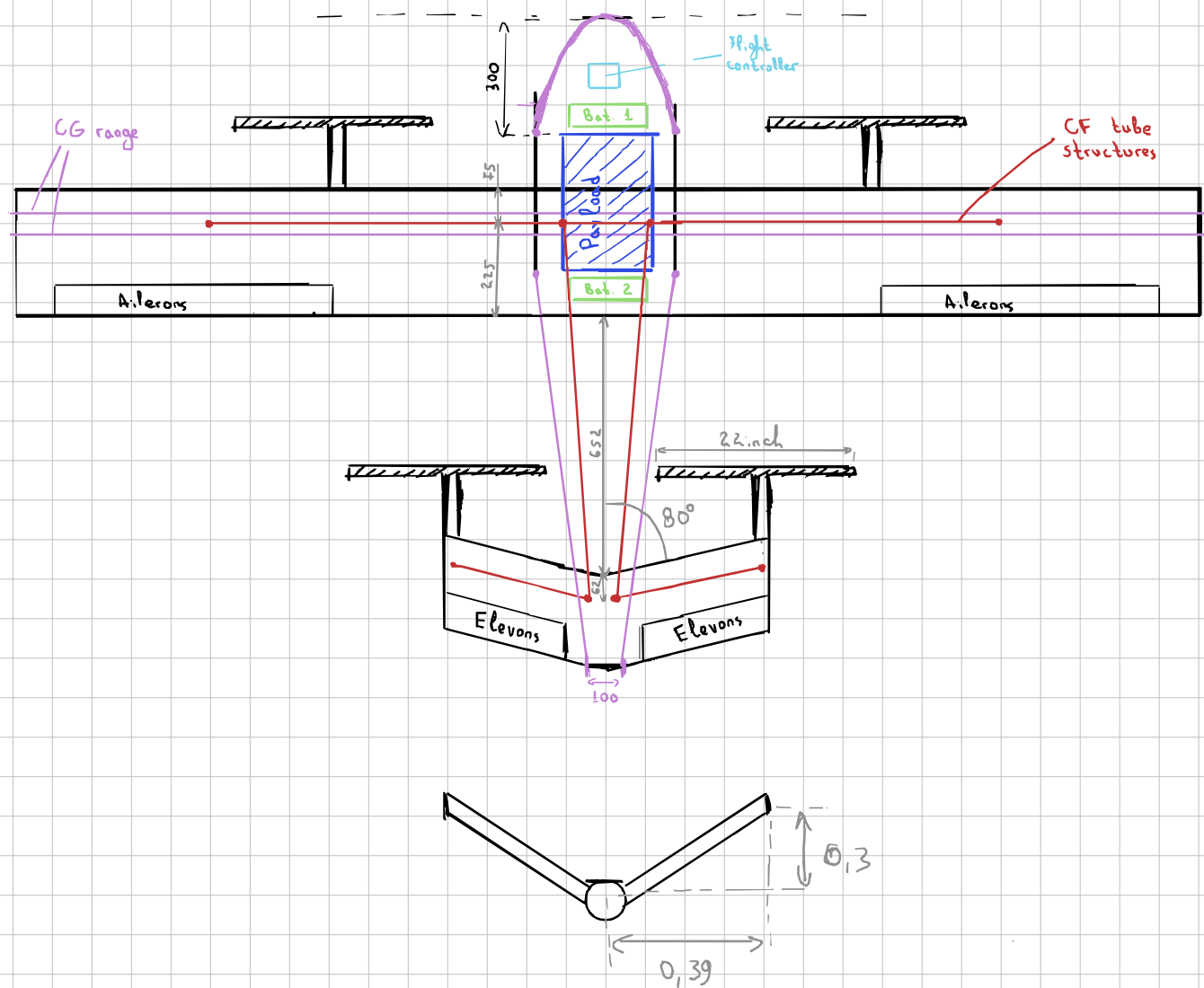
$$\bar{X}_{AC,w} = \frac{X_{0,w} + \overset{\%}{(n_{AC,w} \cdot C_{root,w})}}{C_{MAC}}$$

$$C_{Ld,w} = \frac{2\pi \cdot AR}{2 + \sqrt{\frac{AR^2 \beta^2}{k^2} \cdot (1 + 0)} + 4}$$

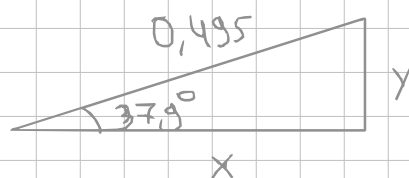
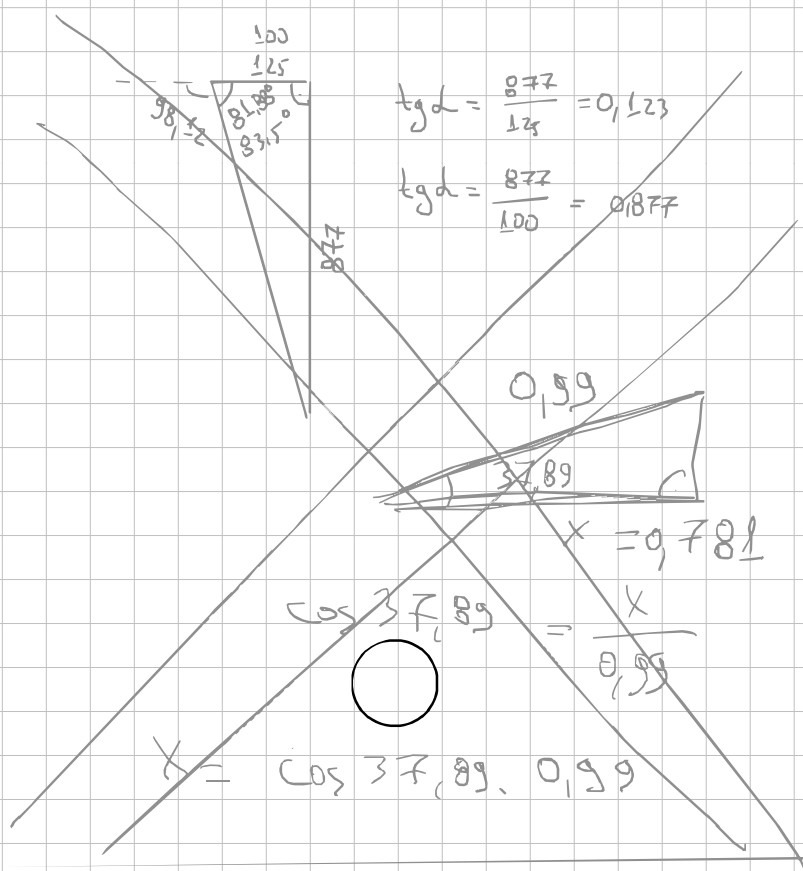
09.04.2026



Last Update : 01. May. 2026



Not correct!



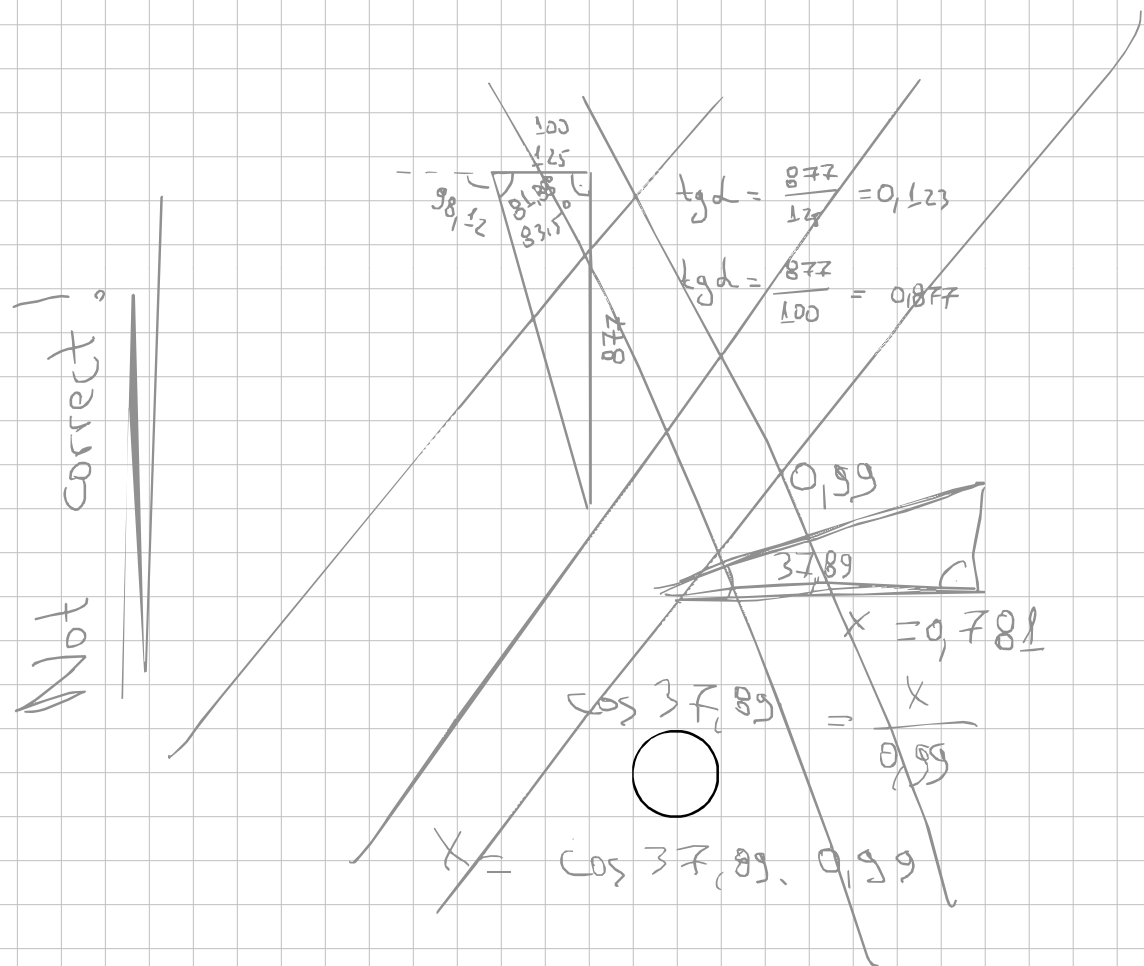
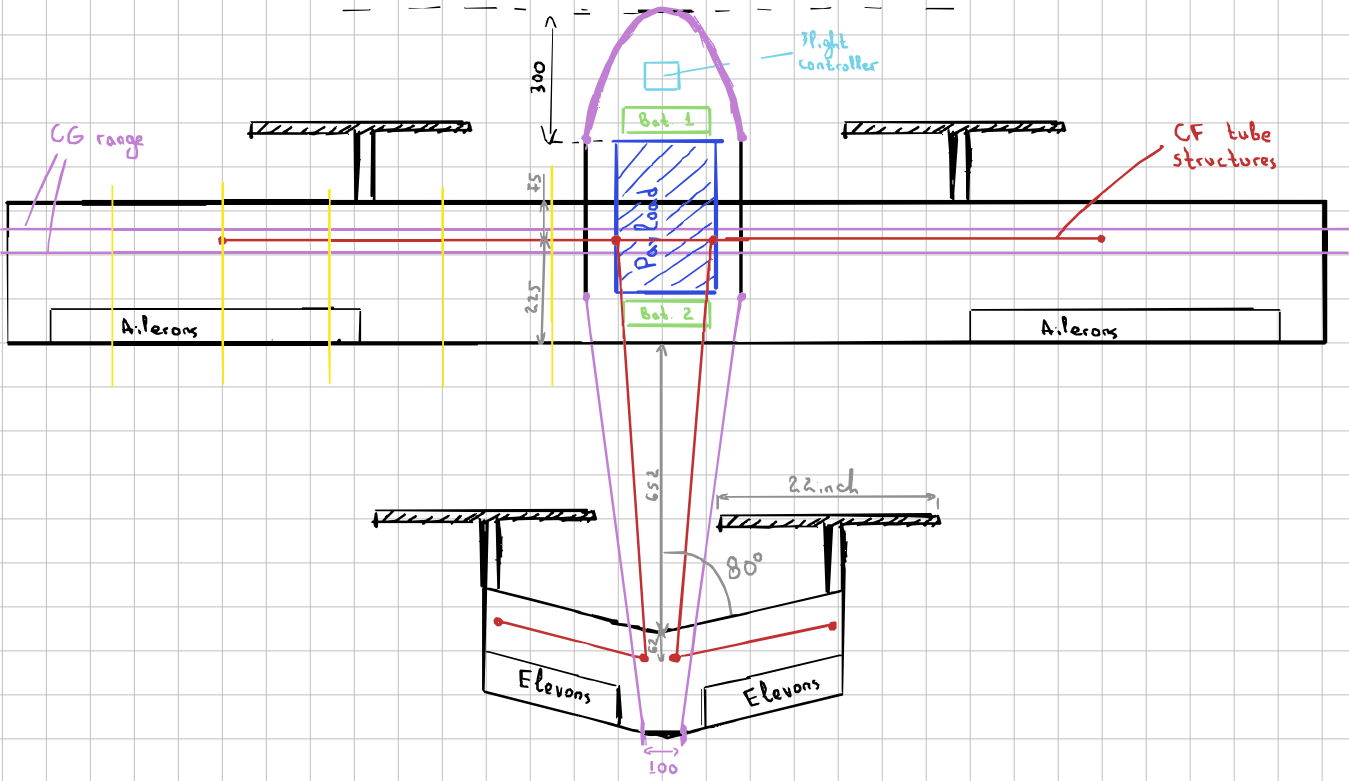
$$\cos 37.9 = \frac{x}{0.495}$$

$$x = 0.789 \cdot 0.495 = \underline{\underline{0.39}}$$

$$\sin 37.9 = \frac{y}{0.495}$$

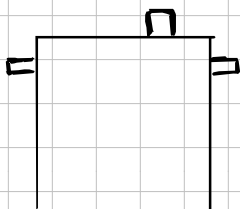
$$y = \underline{\underline{0.304}}$$

Last Update: 01.05.2026



13.04.2026

Tilt Mechanism Concept:



01.05.2026

